EVEREST ENGINEERING COLLEGE

(Affiliated to Pokhara University)

Sanepa - 02, Lalitpur



ASSIGNMENT

OF

SOFTWARE ENGINEERING

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Question 1:

The exam control office of Pokhara university aims to digitize its services, making all required tasks accessible online. The standard operating procedures for each task are already well defined. While the project has a reasonable timeline for completion, it must be executed within a limited budget.

Which software process model would be more suitable for developing this software? Analyze and discuss the advantage of the chosen process model.

Answer:

To digitize the exam control office of Pokhara university, the standard operating procedures are well-defined and the project has a fixed timeline and also limited budget, so the **Waterfall Model** would be the most suitable software process model.

Advantages of the Waterfall Model To digitize the exam control office of Pokhara university Project are:

Clear Structure: The Waterfall model provides a clear, linear structure for the project, making it easy to understand and manage, And also each task is already well defined.

Predictable Timeline: The sequential nature of the Waterfall model allows for a predictable timeline, helping to ensure that the project is completed within the given timeframe. And also a reasonable timeline for completion also matters for this project.

Well-Defined Requirements: Since the standard operating procedures are already well-defined, the Waterfall model can effectively translate these requirements into specific project phases.

Budget Control: To complete this project a limited budget frame is available hence, The phased approach of Waterfall enables better budget control.

Suitable for Stable Requirements: as we know the frames and standards are already fixed there might not be the frequent changes and iterations so The Waterfall model is best for projects.

Hence, To digitize the exam control office of Pokhara university the Waterfall model is a suitable choice due to its well-defined requirements, fixed timeline and budget.

Question 2:

A Flight Control Software System is a specialized, safety- critical software designed to manage and regulate an aircraft's flight control surfaces, such as ailerons, rudders, elevators, and flaps. Its primary purpose is to ensure stability, maneuverability, and safety during all phases of flight, including takeoff, cruising, and landing.

Key Functions:

1. Automatic Flight Control: Supports autopilot operations to maintain altitude, heading, and speed.
2. Manual Flight Assistance: Enhances pilot control by interpreting input from the cockpit controls and translating them into precise movements of the flight surfaces.
3. Flight Envelope Protection: Ensures the aircraft operates within safe limits by preventing dangerous maneuvers, such as stalling or exceeding structural stress thresholds.
4. Sensor Integration: Gathers data from sensors (e.g., airspeed, altitude, and gyroscopic data) to dynamically adjust control actions.
5. Redundancy and Fault Tolerance: Incorporates multiple levels of redundancy to continue functioning in case of component or system failure.

Importance:

- Flight control software systems are critical for modern fly-by-wire aircraft, where traditional mechanical and hydraulic controls are replaced or augmented by electronic systems. Their performance and reliability are essential to ensure passenger safety and compliance with stringent aviation standards like DO- 178C (Software Considerations in Airborne Systems and Equipment Certification)

Which software process model would be most suitable for developing this software? Analyze and discuss the strengths and weaknesses of the chosen process model.

Answer:

Given the critical nature of flight control software, the Spiral Model would be the most suitable software process model. This Spiral Model is best used when dealing with projects that involve a high degree of complexity, accompanied by high risk and evolving requirements like managing and regulating an aircraft's flight control surfaces, such as ailerons, rudders, elevators, and flaps. Its primary purpose is to ensure stability, maneuverability, and safety during all phases of flight, including takeoff, cruising, and landing.

Question 3:

Imagine developing a smart tourist guide system that provides end-to-end services for tourists. This will be a public application accessible to all users.The goal is to create engaging, value-added features while building a community and ensuring sustainability through revenue generation. However, the specific features that will attract and benefit users are yet to be identified.

Which software process model would be more suitable for developing this software? Analyze and discuss the strength and weakness of the chosen process model.

Answer:

The Agile model is one of the best development processes for a smart tourist guide system because it offers considerable flexibility and can easily adhere to user needs. Because not all the particular features of the application have been determined, Agile allows for further corrections as new ideas and needs appear during development. Its iterative nature lets the development team quickly produce an MVP, focusing only on basic functionalities such as navigation or local insights, with a gradual addition of more complex features like gamification and community forums. With Agile, continuous user feedback is encouraged in order to make the application evolve toward the needs of tourists and stakeholders. Also, the revenue generation features can be iterated and tested for premium services or advertisements that will keep the app sustainable.